Code No: 134AK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, February - 2024
COMPUTER ORGANIZATION
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) What is computer organization? [2]
b) What is the significance of AC, IR and PC registers? [3]
c) What is the maximum memory size that can be addressed by 8086? [2]
d) Briefly explain the four segment registers of 8086 processor. [3]
e) What is an assembler? [2]
f) What is the purpose of DEBUG command? List out the function of various command characters of DEBUG command. [3]
g) State the basic principle of the two-wire handshaking method of data transfer. [2]
h) Draw the timing diagram for asynchronous data transfer. [3]
i) What is Auxiliary memory? [2]
j) Explain the need and main motivation behind virtual memory. [3]

PART – B

(50 Marks)

2.a) Present an initial configuration for the instruction cycle using a flow chart and explain how the control determines the instruction type after decoding phase?
b) List out various phases of an instruction cycle. [7+3]

OR

3.a) Explain the significance of micro programmed control unit.
b) Draw the block diagram of a control memory and its associated hardware needed for selecting the next microinstruction address and explain. [5+5]

4. Draw the architecture of 8086 processor and describe the overall organization of each unit. [10]

OR

5. List and explain the functions of various data transfer, branch and loop instructions of 8086 processor with examples for each. [10]

6. How to define and call a subroutine in 8086 programming? Explain with an illustrative example. [10]

OR

7.a) Explain the interrupt response sequence of 8086 when an interrupt occurs.

b) Write an assembly language program to add the contents of the memory location 2000H:0500H to contents of 3000H:0600H and store the result in 5000H:0700H. [6+4]

8. Draw the flowchart and explain the algorithm for adding and subtracting two binary numbers in signed 2's complement representation. [10]

OR

9.a) Differentiate between synchronous and asynchronous data transfer.

b) Explain daisy-chaining priority interrupt method for data transfer with relevant diagrams. [4+6]

10. Draw the block diagram of associative memory and explain its functionality in detail. [10]

OR

11. What are the conflicts raised during multiple processes communication? Explain various ways to implement interprocessor synchronization? [10]

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